

Your submitted solution is only **valid** if it contains the following paragraphs at the top of the program. Please *fill* in *your name* and *Neptun code*.

```
/*<NAME>, <NEPTUN>
```

```
This solution was prepared and submitted by the student stated above for the assignment of the Programming course. I declare that this solution is my own work. I have not copied or used third party solutions. I have not passed my solution to my classmates, neither made it public.
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Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§) states that as long as a student presents another student's work - or at least the significant part of it - as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a disciplinary fault can be dismissal of the student from the University.*//
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When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: if you are solving a) do not include the outputs for tasks b), c), d)!



Enter the Schengen zone

There are n lanes at the cross-border station of the EU Schengen zone, and cars are waiting to transit in m positions at each lane. Every person's passport must be scanned before entering the zone, and each scan takes 30 seconds.

An $n \times m$ matrix stores the number of passengers per position. If the number is 0, there is no car in the given position (generally at the end of lanes)

Input

- The first line of input contains n ($2 \leq n \leq 100$), the total number of lanes, and m ($2 \leq m \leq 100$), the number of car positions in each lane.
- Each of the next n lines contains p ($0 \leq p \leq 9$), the number of passengers in the given car in the given position. If a value is 0, it means there is no car in the given position. In this case the further positions are also blank from it in the given lane.

Write a program that solves the following tasks:

Tasks:

- a. How many cars are waiting to cross?
- b. Display the index of the lane, where the sum of the required time for passport scanning in the smallest.
- c. Is there any lane where the number of passengers in cars strictly increasing (if $i > j$, then $\text{car}[i] > \text{car}[j]$).
- d. Select the lane with the fewest single-occupant cars waiting. If there is no lane with at least one single-occupant car waiting, print -1.

Output

Each task's output should be separated by new lines.

Test Case 1

Input	Output
5 7	32
2 2 3 4 1 7 0	4
1 2 3 4 5 6 7	True
7 4 3 3 3 0 0	1
1 1 2 2 3 4 5	
2 2 2 3 3 3 5	

Test Case 2

Input	Output
3 3	7
2 2 2	2
1 0 0	False
2 4 1	2

Evaluation based on 10 input tester files:

Points: $10 \cdot (2+2+3+3) = 10 \cdot 10 = 100$ points